

Strings & StringBuilder

def: A string stores a sequence of characters(text). String builder is a mutable version used when you need to modify text frequently & efficiently.

How to use: String name = "shemmy";

S.o.p(name.length()); // 6

S.o.p(name.toUpperCase()); // SHEMA

StringBuilder sb = new StringBuilder("Hello");

sb.append(" world");

S.o.p(sb); // Hello World

when to use: string for fixed or rarely-changing text.
StringBuilder when you're modifying/building text repeatedly (e.g., inside a loop) - much faster than repeatedly creating new strings.

where to use: Text processing, user names, messages, building output dynamically.

why StringBuilder exists separately from Strings: Strings

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Date: / /

in Java are immutable (can't be changed once created) - every "modification" actually creates a brand new string, which is slow if done repeatedly. `StringBuilder` solves this by allowing in-place changes, making it much faster for heavy text manipulation.

Real-life ex: `String` is like a printed document - to change even one word, you repaint the whole page. `StringBuilder` is like a whiteboard - you can erase & rewrite parts without recreating the whole thing.